5.9. SECONDARY PROCESSES IN THE ATOM

Once an electron is ejected from an atomic orbital due to internal conversion, electron capture, or some other process involved in radioactive decay, a vacancy is created in the electron shell which can be filled in several ways. Electrons from higher energy orbitals can occupy the vacancy. The difference in the binding energy of the two shells involved in the transition will be emitted from the atom as *X-rays*. This process is called *fluorescent radiation*.

If the difference in the binding energy for the transition is sufficient to exceed the binding energy of electrons in the L- or M-levels, emission of the energy as X-rays is not the predominant mode. Instead an *internal photoelectric* process can occur and the binding energy results in the emission of several low energy electrons which are called *Auger electrons*. Auger electrons are much lower in energy than the electron from the *nuclear internal conversion* process, since the difference in electronic binding energies is in the eV range compared to the energies in the nuclear conversion process which are in the MeV range. The atom may be left in a state of high ionization by Auger emission; positive charges of 10−20 have been observed. When such high charges are neutralized, the energy liberated is sufficient to break chemical bonds.

In isomeric decay the γ-energy is often so small that the daughter recoil is negligible. For example

$$^{80m}\mathrm{Br} \xrightarrow[4.4\ \mathrm{h}]{\gamma} {}^{80}\mathrm{Br} \tag{5.53}$$

occur through the emission of a γ-ray of 0.049 MeV, giving the daughter a recoil energy of only 0.016 eV. Still the decay leads to the emission of ^{80}Br from ethyl bromide, when the parent compound is C_2H_5 ^{80m}Br, even though the bond strength is 2.3 eV. This is because the γ is highly converted and as the electron "hole" is filled, Auger emission occurs. Bromine ions from Br^+ to Br^{17+} have been observed through mass spectrometric analyses of the ethyl bromide gas phase.

5.10. CLOSED DECAY ENERGY CYCLES

The masses for many short-lived nuclei are unknown although their decay modes and energies have been determined. From this the nuclear masses may be calculated, and consequently *Q*-values of different unknown decay modes can be obtained. This can be done through the use of closed decay energy cycles.

Suppose we need to know if ^{237}U can decay to ^{233}Th through α-emission. Of course this is a simple calculation if the masses of ^{237}U and ^{233}Th are known, but let us assume they are not. We have data that ^{237}U decays through β-emission (E_{max} 0.248 MeV) followed by γ-decay (E_γ 0.267 MeV). ^{233}Th decays through β-emission (E_{max} 1.230 MeV) directly to ^{233}Pa. ^{237}Np undergoes α-decay to ^{233}Pa with $E_\alpha = 4.79$ MeV. We may construct a closed cycle including these decay energies as shown in Figure 5.7.C.

The Q-value for branch D is; $Q = -931.5\ (M_{233\mathrm{Th}} + M_{\mathrm{He}} - M_{237\mathrm{U}})$. For branch A we can calculate (5.18); $Q = E_\alpha\ M_Z\ /M_{Z-2} = 4.79 \times 237/233 = -931.5\ (M_{233\mathrm{Pa}} + M_{\mathrm{He}} - M_{237\mathrm{Np}})$. By introducing values for M_{He} and $M_{237\mathrm{Np}}$ we obtain $M_{233\mathrm{Pa}} = 233.040\ 108$. For branch B we calculate; $M_{233\mathrm{Th}} = M_{233\mathrm{Pa}} + 1.230/931.5 = 233.041\ 428$. For branch C one obtains; $M_{237\mathrm{U}} = M_{237\mathrm{Np}} + (0.248 + 0.267)/931.5 = 237.048\ 581$.

Thus all information is available for calculating branch D. The Q-value is found to be 4.23 MeV, and the $E_\alpha = 4.23 \times 233/237$ or 4.16 MeV. Although spontaneous α-decay is energetically possible, it has not been detected. The systematics of α-decay (§5.17) indicates an expected half-life of $>10^6$ y. Because the β-decay rate is much faster ($t_{½} = 6.75$ d), too few α's are emitted during the life-time of ^{237}U to be detected.

5.11. KINETICS OF SIMPLE RADIOACTIVE DECAY

Most radioactive isotopes which are found in the elements on earth must have existed for at least as long as the earth. The nonexistence in nature of elements with atomic numbers greater than 92 is explained by the fact that all the isotopes of these elements have life-times considerably shorter than the age of the earth.

Radioactive decay is a random process. Among the atoms in a sample undergoing decay it is not possible to identify which specific atom will be the next to decay. Let us make the following assumptions for a nucleus to undergo decay during a time interval Δt.

1. The probablity is independent on the history of the nucleus (i.e. equal for all t)
2. The probablity is independent on the surroundings of the nucleus
3. The probability is equal for all nuclei of a certain type and exitation energy

There are some cases where "2" is not valid but for the vast majority of cases the assumptions 1-3 are valid. Now, consider a selected nucleus and let s be the probability of decay during Δt. However, s increases proportionally to Δt according to 1. Let also p be the probability for not decaying during Δt. Then, $s + p = 1 \rightarrow p = 1 - s$ but $s = \lambda \Delta t$ and therefore $p = 1 - \lambda \Delta t$. Now, let q be the probability for not decaying during the time $t = \Sigma \Delta t$. (with $q = \Pi p$ and $p = t/\Delta t$). Since all p are equal for equal Δt one gets:

$$q = (1 - \lambda \Delta t)^{t/\Delta t} = \lim_{\Delta t \to 0} \left((1 - \lambda \Delta t)^{-1/(\lambda \Delta t)} \right)^{-\lambda t} = \mathrm{e}^{-\lambda t} \tag{5.54}$$

For a large number of atoms (>100) probability may be approximated by frequency, why (according to 3 are all q equal): $N/N_0 = \mathrm{e}^{-\lambda t} \rightarrow N = N_0 \mathrm{e}^{-\lambda t}$, or $\ln(N/N_0) = -\lambda t$

Taking the derivative with respect to t yields:

$$\mathrm{d}N/\mathrm{d}t = -\lambda N \tag{5.55}$$

where λ is the proportionality constant known as the *decay constant*.